What we're doing

Score the new MarinFold 1.5B checkpoint (`protein-contacts-1_5b-distance-masked-70f8f5/step-49999`) on the 100 FoldBench monomer subset already collected by exp12 / exp20. Compare head-to-head against three baselines on the same proteins with the same distogram-derived metrics:

- `marinfold_1b` — exp20's 1B checkpoint (the model we're trying to outscale) - `protenix_single_seq` — Protenix v2 in single-sequence mode - `protenix_msa` — Protenix v2 with MSA conditioning (gold standard)

Inference: zero-shot, CB-CB pair sweep with vLLM prefix cache, one trunk + $N^2/2$ tails per protein. Atom convention matches Protenix's distogram representative atom (CB with CA fallback for GLY / UNK).

Why

Direct rerun of exp20 with the new 1.5B checkpoint substituted for 1B. Question being asked: does scaling from 1B to 1.5B at the current training budget translate into better zero-shot distogram quality on FoldBench monomers — specifically, does it close the gap to Protenix-single-seq?

Pre-registered hypothesis (issue #26): 1.5B beats 1B on at least **3 of 4** headline metrics — `lddt_distogram_cb` (↑), `mae_distogram_cb_angstrom` (↓), `drmsd_distogram_cb_angstrom` (↓), `prec_long_L` (↑). The bar is intentionally low.

Compute

iris on TRC, v5p-8 (matches the marin protein-eval convention; the protein-training-1b branch in marin uses the same `--extra vllm --extra tpu` launch shape). Full 100-protein run finished in a single ~4 h job. Outputs land per-protein on GCS at `gs://marin-us-east5/protein-structure/MarinFold/exp26/protein-contacts-1_5b-distance-masked-70f8f5-step-49999-foldbench-monomers/<stem>/{distogram.npz, provenance.json}` so partial progress would survive preemption.

vllm-tpu version pin worth flagging: we had to inline marin's `vllm` extra (vllm-tpu 0.19 / tpu-inference 0.19 / libtpu 0.0.39 as published in `marin-latest`) rather than pulling `marin[vllm]` transitively — uv source maps don't propagate the pytorch-cpu index registration through transitive extras, so the `torchvision==0.25.0+cpu` pin couldn't resolve.

Headline result

Hypothesis **not supported (2 / 4 metrics).**

| metric | 1.5B | 1B | Δ vs 1B | direction | |---|---:|---:|---:|---| | `lddt_distogram_cb` ↑ | **0.288** | 0.272 | +0.016 | ✓ support | | `prec_long_L` ↑ | **0.285** | 0.258 | +0.026 | ✓ support | | `mae_disto_cb` (Å) ↓ | **5.695** | 5.642 | +0.053 | ✗ refute | | `drmsd_disto_cb` (Å) ↓ | **7.202** | 7.143 | +0.059 | ✗ refute |

The aggregate Δs are small in both directions — neither model is meaningfully moving on this benchmark at this scale.

Per-protein head-to-head

The aggregate hides a sharper picture: 1.5B wins on the majority of proteins for bin-pooled metrics (LDDT, contact precision) but loses on most for per-pair Å-distance metrics (MAE, dRMSD).

metric	1.5B beats 1B	1.5B beats Protenix-SS	1.5B beats Protenix-MSA
`lddt_distogram_cb` ↑	74 / 100	11 / 100	0 / 100
`prec_long_L` ↑	70 / 100	33 / 100	1 / 100
`mae_disto_cb` ↓	40 / 100	6 / 100	0 / 100
`drmsd_disto_cb` ↓	38 / 100	6 / 100	0 / 100

Across all 400 (protein × metric) cells vs Protenix-MSA, MarinFold 1.5B wins on exactly one.

Inference cost

Per-protein wall-time scales $O(N^2)$ (the pair sweep) for both MarinFold models on log-log. Apples-to-apples comparison is muddled by hardware: 1.5B ran on TPU v5p-8 via iris, 1B on H100 via Modal (exp20), Protenix on H100 (exp12). At equivalent N, the 1.5B-on-TPU pair sweep is roughly an order of magnitude slower per protein than 1B-on-H100 — this is a hardware mix story, not an algorithmic regression. Protenix sits roughly flat at ~100 s/protein (trunk-dominated).

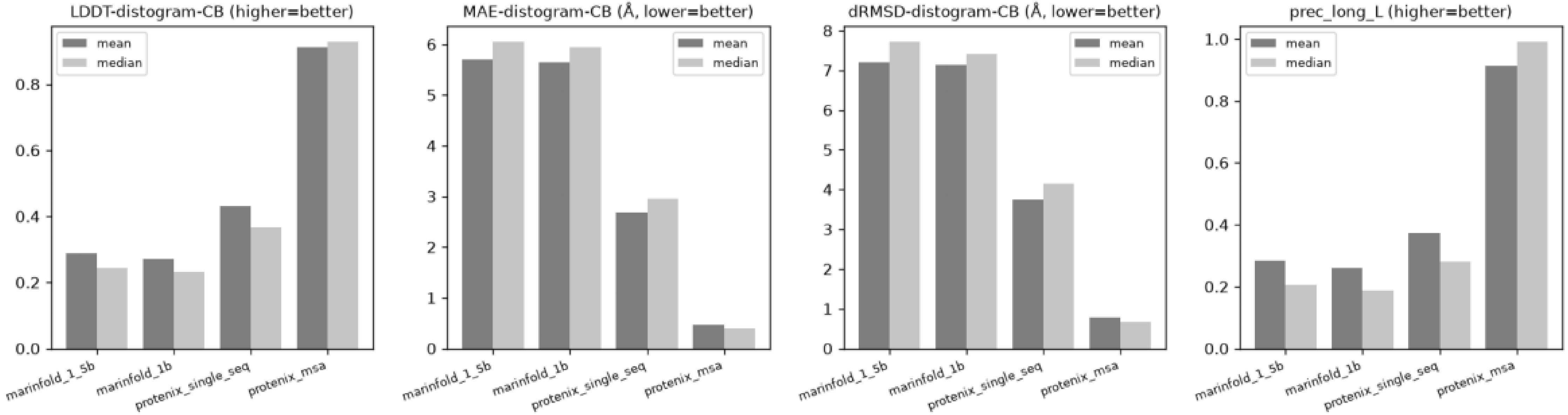
Conclusion

Did 1.5B move us closer to Protenix? Slightly, in the same shape 1B already had. Small bump on bin-pooled contact-class metrics (LDDT +1.6 pp, prec_long_L +2.6 pp), no movement on the metrics you'd actually use for downstream structure work (per-pair Å error). Gap to Protenix-single-seq stays wide on every metric (LDDT 0.288 vs 0.432); gap to Protenix-MSA is ≈3×.

Headline read: scaling 1B → 1.5B at the current training budget is **not the lever that closes the gap**. Natural next move is on the training-side levers (more tokens, distance loss reweighting, MSA conditioning) rather than further parameter scaling alone.

headline_aggregate.png

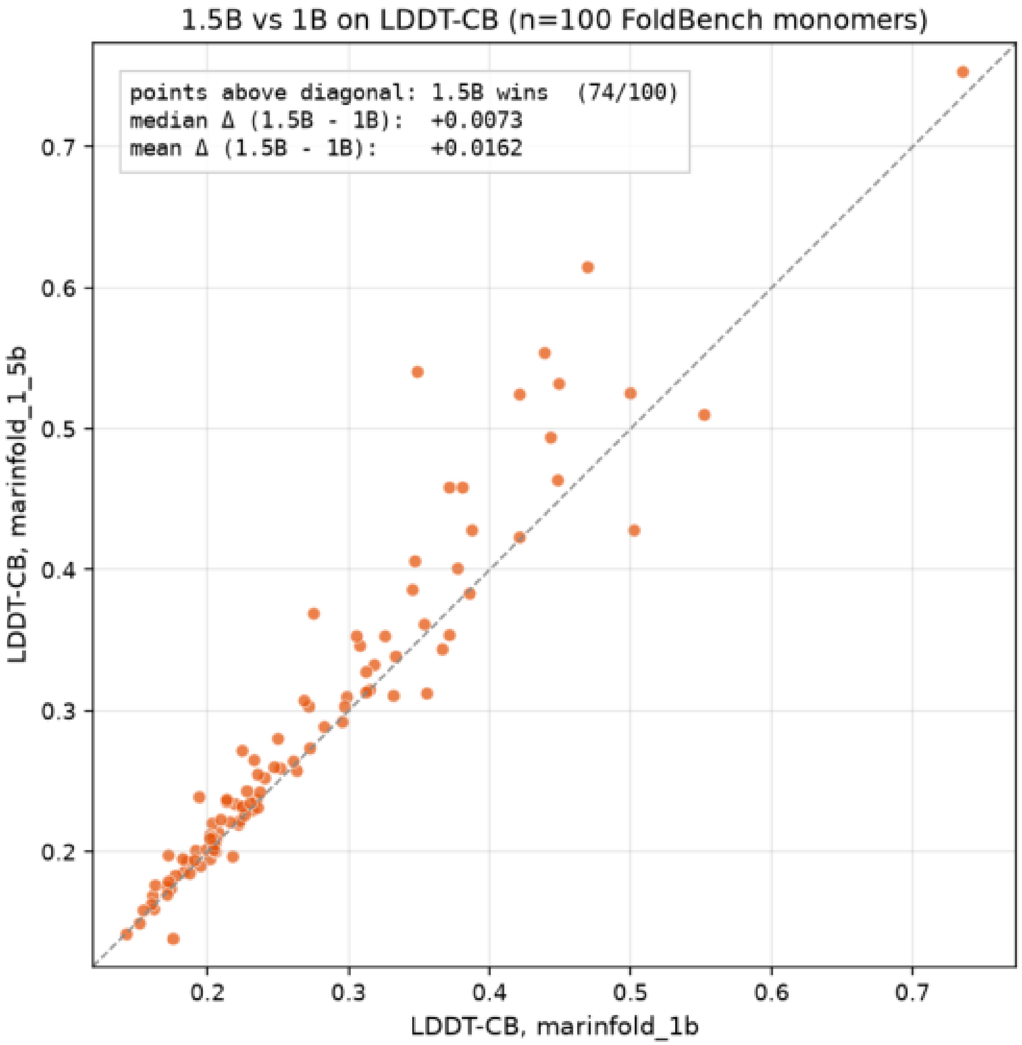
Aggregate (mean + median) for the 4 headline metrics × 4 methods. Hypothesis verdict (1.5B vs 1B, 3/4 bar): 2/4 metrics support → not supported.



$ plot_comparison.py

lddt_1_5b_vs_1b_scatter.png

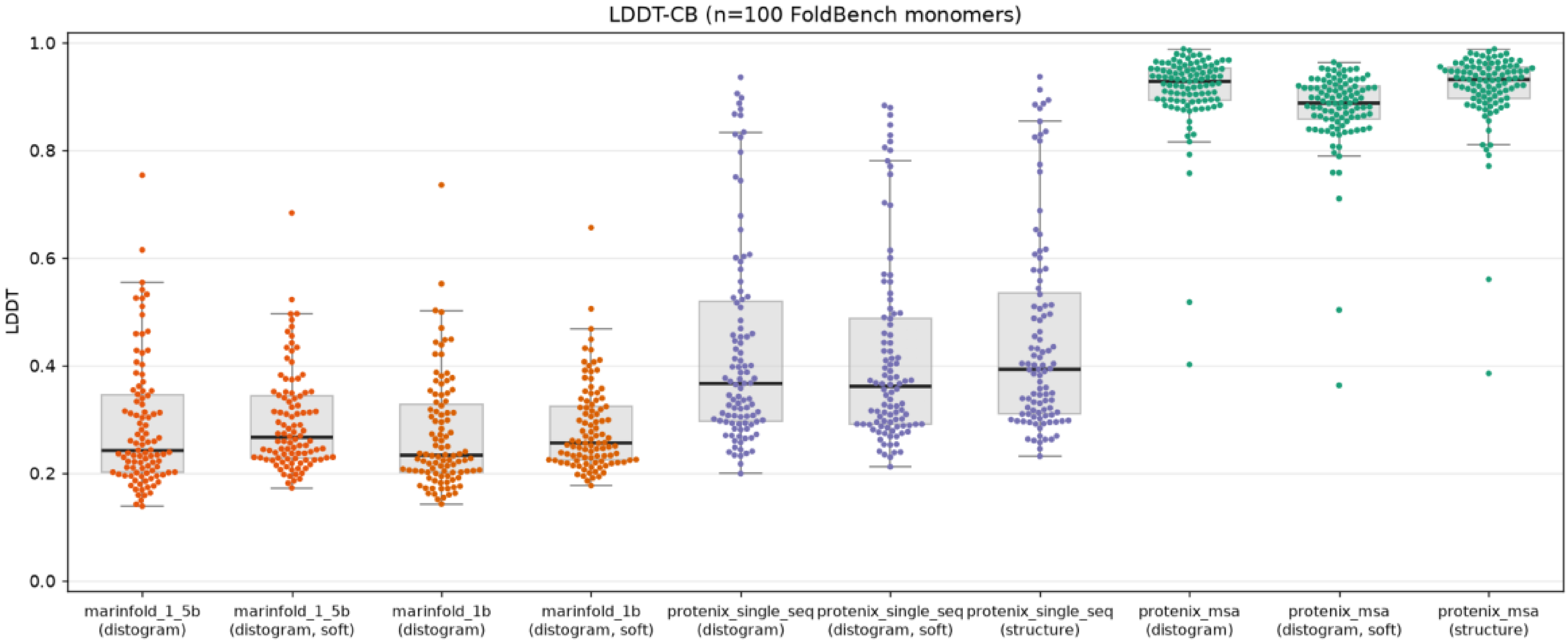
Standalone 1.5B vs 1B LDDT-CB scatter, one point per protein (n=100). Square equal-axes; points above diagonal = 1.5B wins (74/100). Median Δ +0.0073, mean Δ +0.0162.



$ plot_comparison.py

lddt_5way_swarm.png

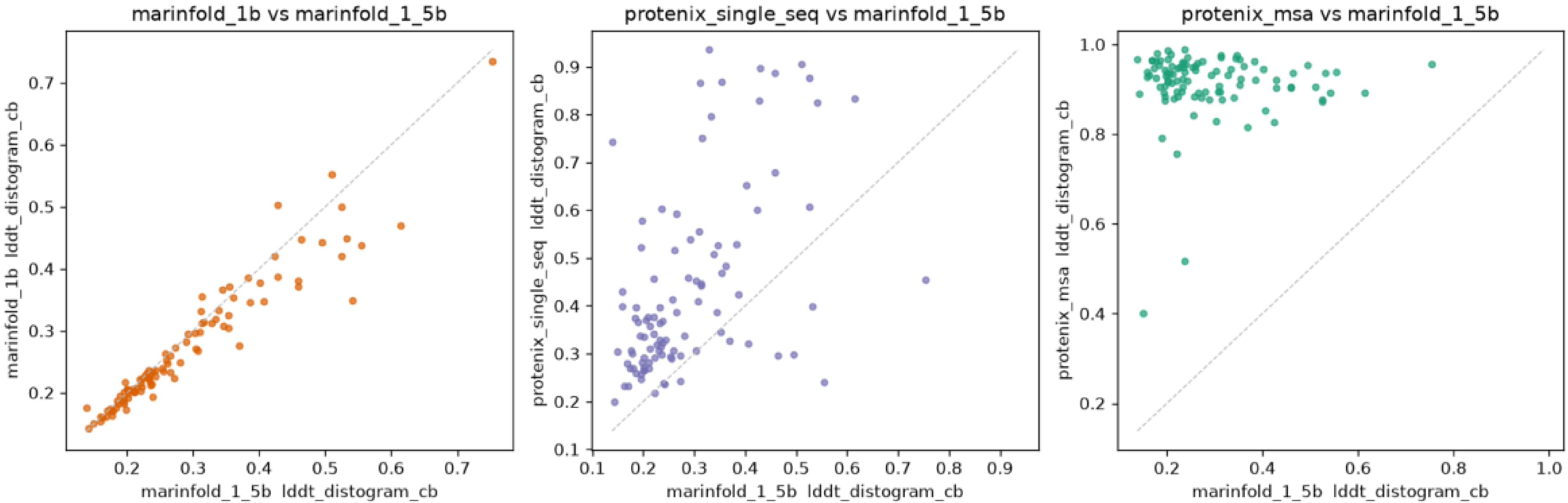
LDDT-CB across 10 categories (4 methods × {distogram, distogram-soft, structure where available}). Boxplot overlay shows median + quartiles. Protenix-MSA dominates ~0.93; MarinFold 1.5B and 1B sit nearly on top of each other at ~0.25.



$ plot_comparison.py

lddt_marinfold_vs_protenix_scatter.png

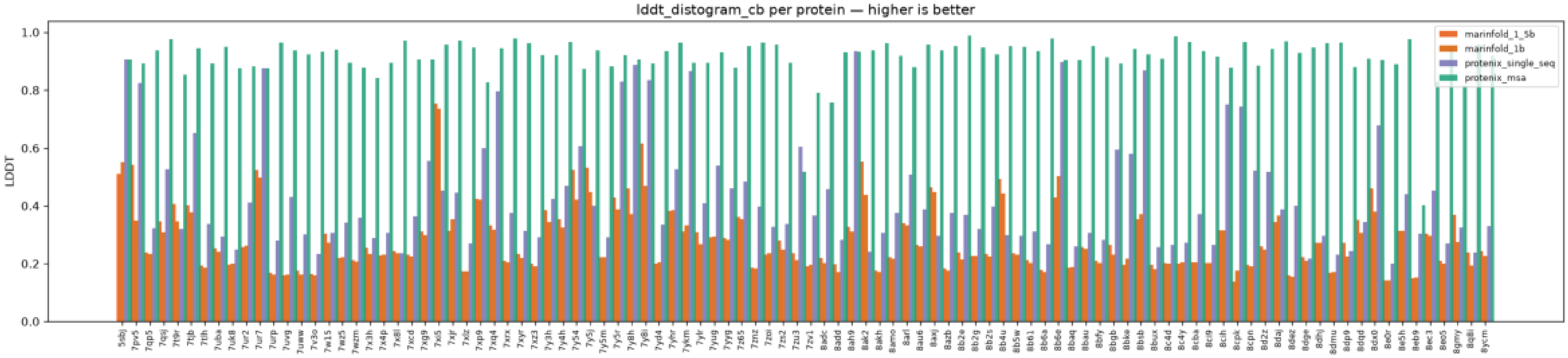
Paired per-protein lddt_distogram_cb scatter, 1.5B on the x-axis. Points above the y=x diagonal = comparator wins. The 1.5B-vs-1B panel is the headline view.



$ plot_comparison.py

lddt_per_protein.png

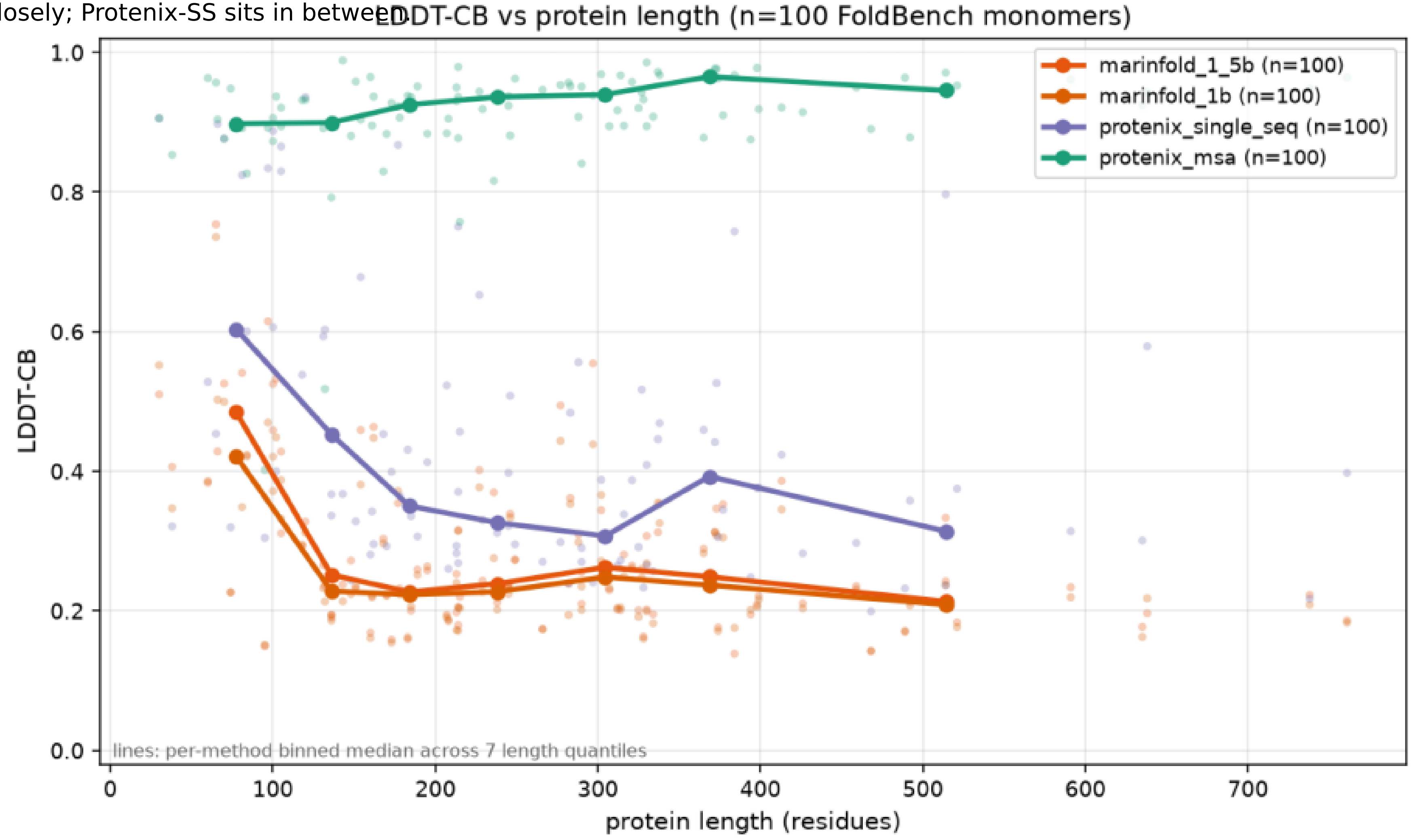
Per-protein lddt_distogram_cb grouped bar across all 4 methods (n=100 FoldBench monomers). Higher is better.



$ plot_comparison.py

lddt_vs_protein_length.png

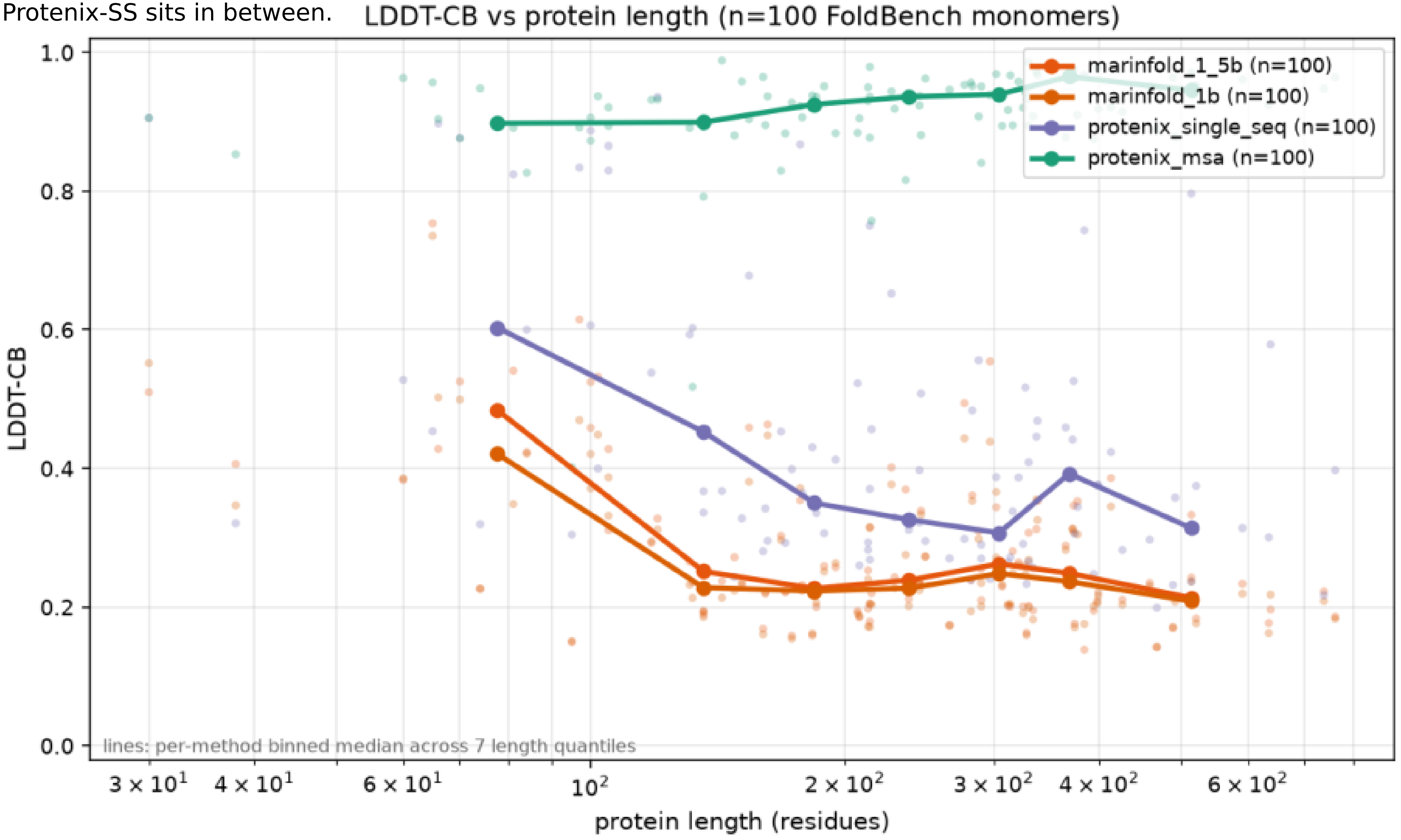
LDDT-CB vs protein length (n_residues, linear x-axis), all 4 methods. Light scatter per protein, lines are per-method binned medians across 7 length quantiles. Protenix-MSA is roughly flat near 0.9; both MarinFold models degrade smoothly with length and track each other closely; Protenix-SS sits in between.



$ plot_comparison.py

lddt_vs_protein_length_log.png

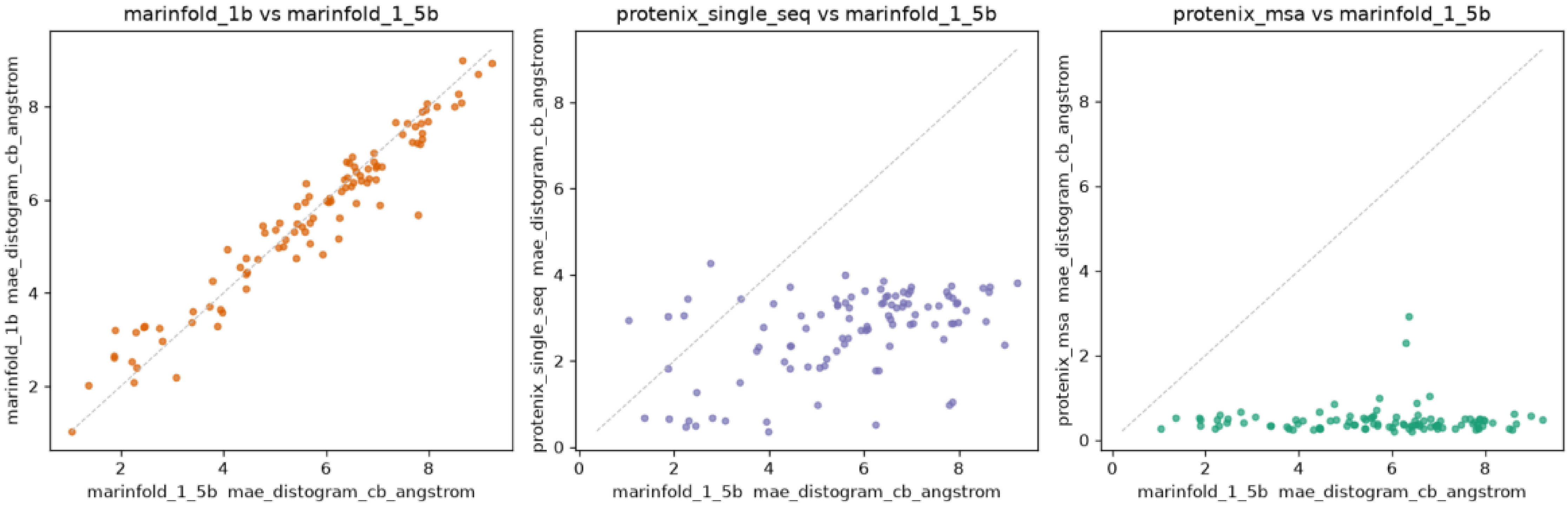
LDDT-CB vs protein length (n_residues, log x-axis), all 4 methods. Light scatter per protein, lines are per-method binned medians across 7 length quantiles. Protenix-MSA is roughly flat near 0.9; both MarinFold models degrade smoothly with length and track each other closely; Protenix-SS sits in between.



$ plot_comparison.py

mae_marinfold_vs_protenix_scatter.png

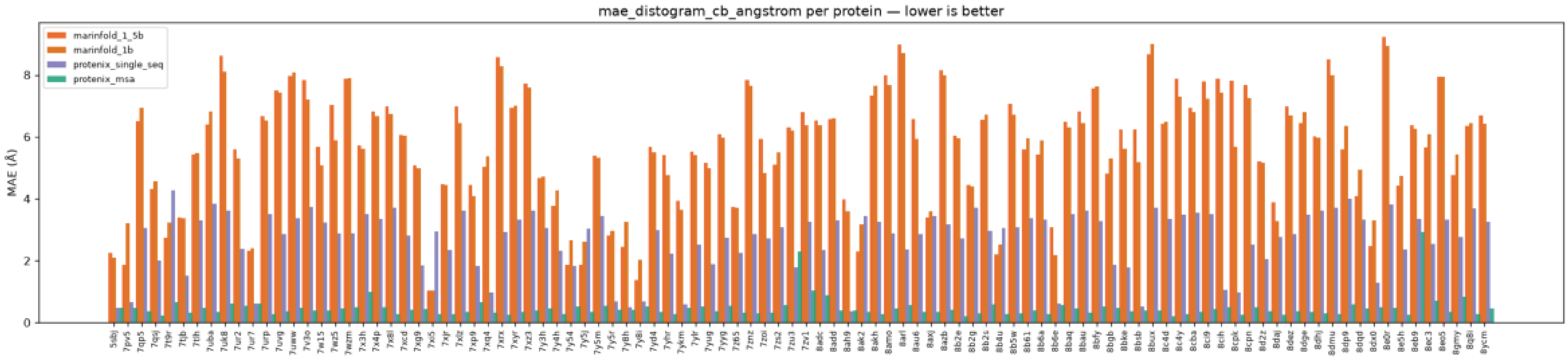
Paired per-protein mae_distogram_cb_angstrom scatter, 1.5B on the x-axis. Points below the y=x diagonal = comparator wins. The 1.5B-vs-1B panel is the headline view.



$ plot_comparison.py

mae_per_protein.png

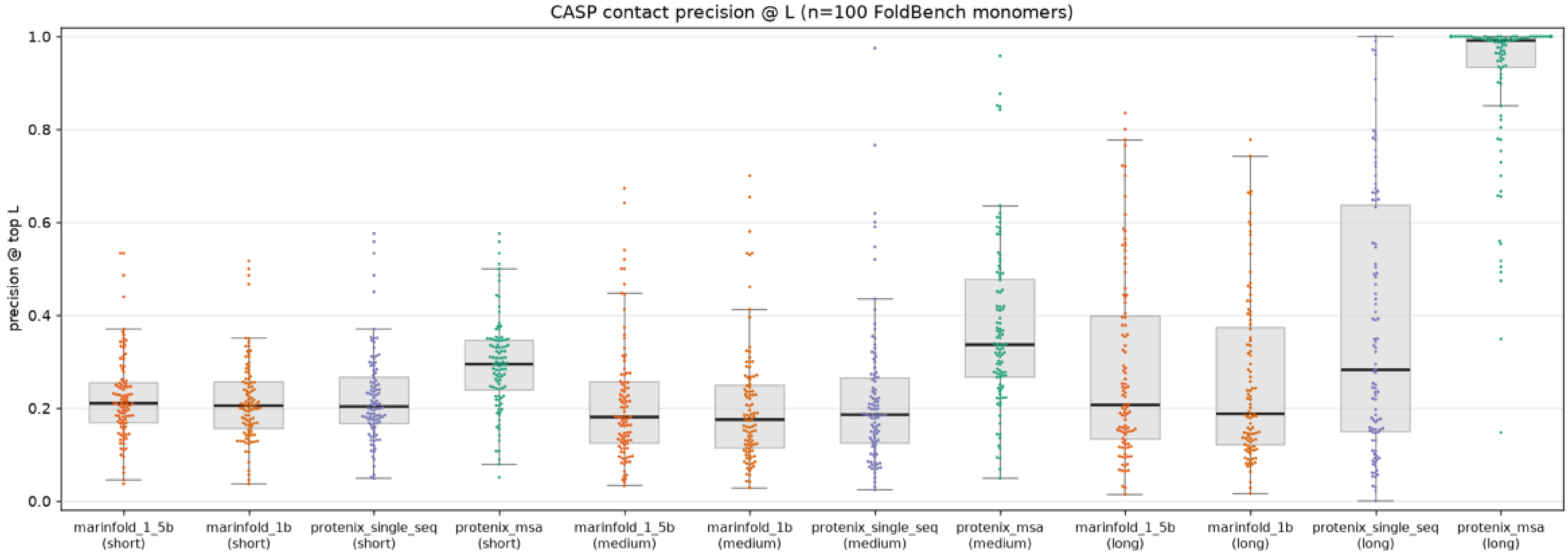
Per-protein mae_distogram_cb_angstrom grouped bar across all 4 methods (n=100 FoldBench monomers). Lower is better.



$ plot_comparison.py

prec_L_swarm.png

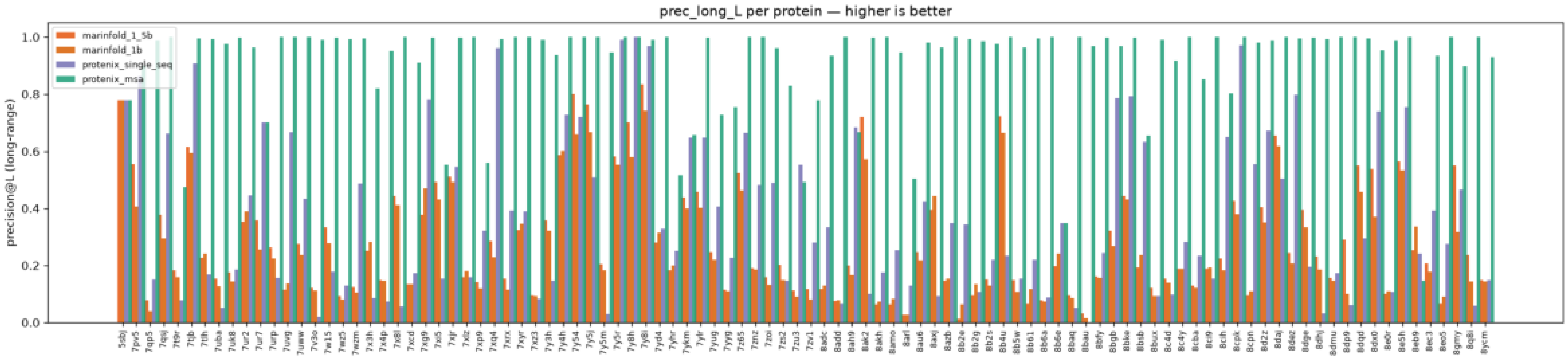
CASP contact precision @ top L, 12 categories (4 methods × {short, medium, long} separation). Long-range is the hardest; 1.5B and 1B sit close to each other on every separation class.



$ plot_comparison.py

prec_long_L_per_protein.png

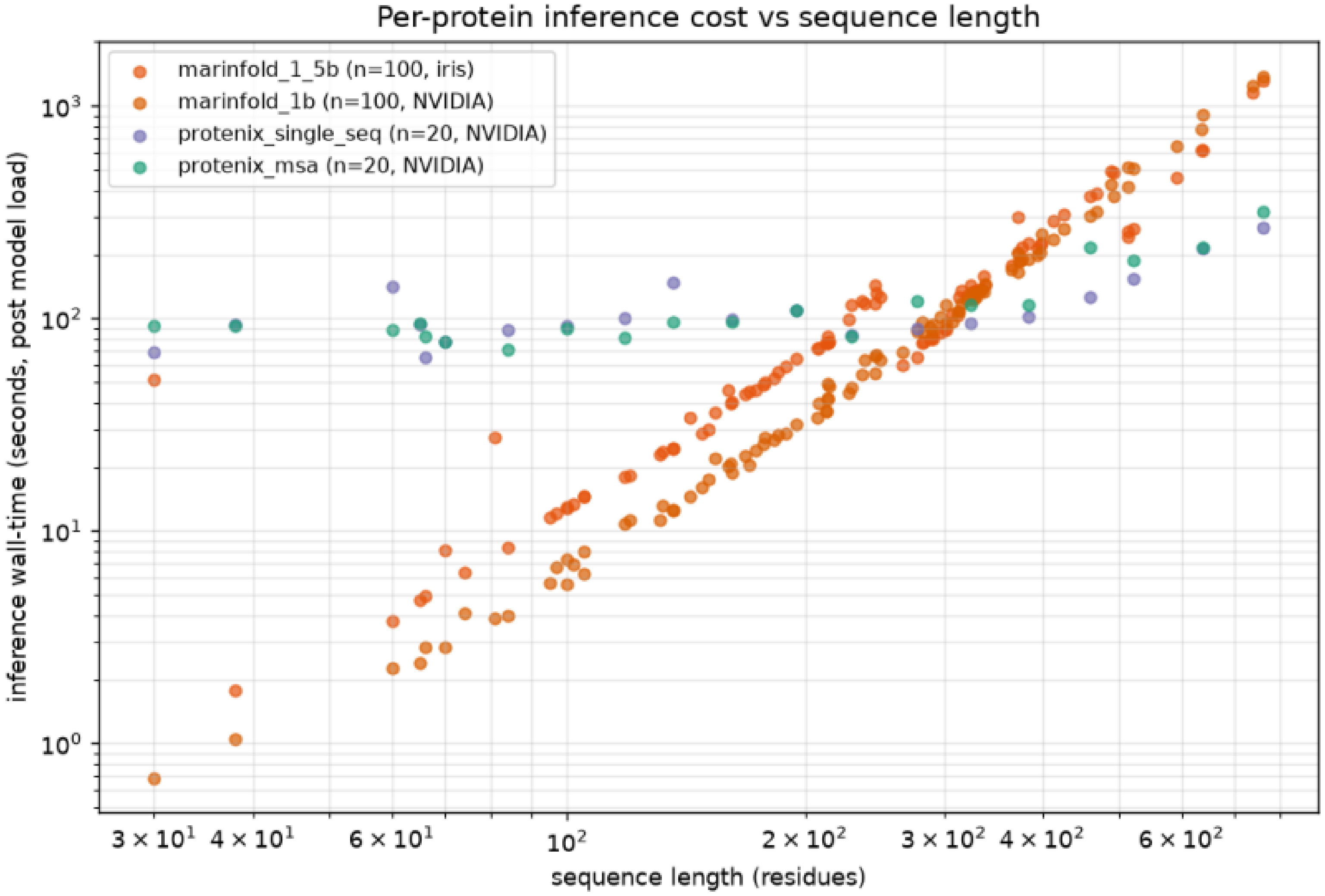
Per-protein prec_long_L grouped bar across all 4 methods (n=100 FoldBench monomers). Higher is better.



$ plot_comparison.py

timing_4way_vs_sequence_length.png

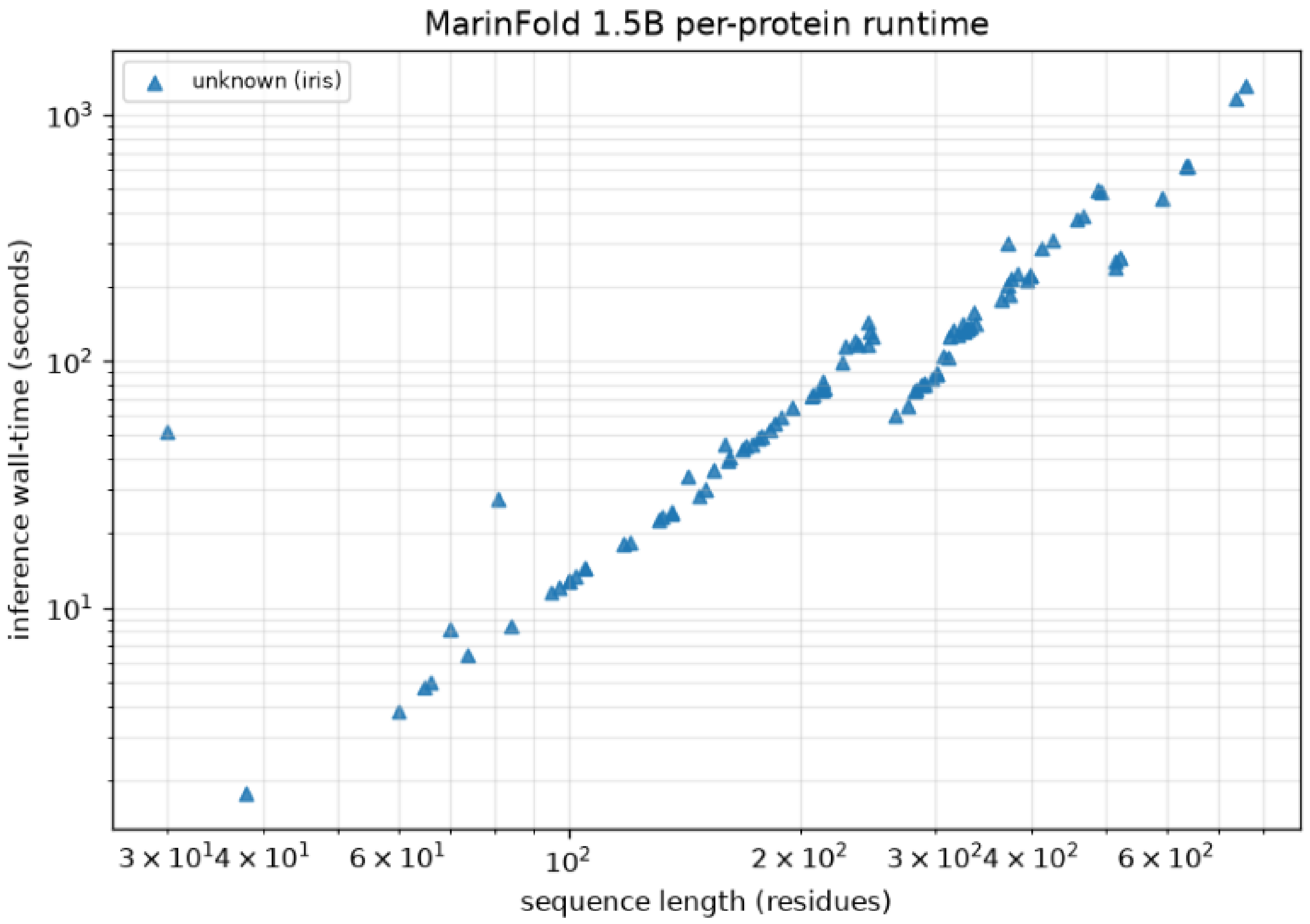
Per-protein wall-time vs sequence length, all 4 methods (log-log). Hardware caveat: 1.5B on TPU v5p-8/iris, 1B on H100/Modal, Protenix on H100. Both MarinFold models show clean O(N²) pair-sweep scaling; Protenix is roughly flat at ~100 s.



$ plot_comparison.py

timing_vs_sequence_length.png

Per-protein wall-time vs sequence length for MarinFold 1.5B (this exp). Log-log; the pair-sweep is $O(N^2)$ so the slope is ~2.



$ plot_comparison.py